

LUCKY SAH

Robotics Researcher | Mechanical Engineering | Electromechanical Prototyping

Kent, OH | lsah@kent.edu | [ADD PHONE] | [ADD LINKEDIN OR PORTFOLIO URL]

PROFESSIONAL SUMMARY

Mechatronics engineering student and robotics researcher who develops electromechanical systems from parametric CAD through fabrication, assembly, integration, and functional hardware. At Kent State University's Advanced Telerobotics Research Lab, leads mechanical engineering and prototyping work while managing engineering resources and project support for interdisciplinary robotics teams. Brings additional industry experience developing modular AI companion hardware and automated production equipment, complemented by international robotics activities and STEM community leadership.

EXPERIENCE

Robotics Researcher

[ADD START DATE] – Present

Advanced Telerobotics Research (ATR) Lab, Kent State University

- Lead mechanical engineering workstreams and prototype development for robotics research, advancing systems from parametric CAD through fabrication, assembly, integration, and functional hardware.
- Own mechanical workstreams spanning structures, mounts, mechanisms, actuator integration, sensor packaging, and embedded hardware integration.
- Manage prototyping resources, equipment, workspace, and hands-on engineering support across interdisciplinary research teams.

Robotics Hardware Intern (Co-op)

[ADD DATES]

PlaiPin Inc.

- Translate parametric CAD concepts into build-ready mechanical assemblies and functional prototypes for a modular AI companion robot.
- Design and iterate structural components, mounts, and actuator interfaces while supporting fabrication, assembly, hardware integration, and prototype refinement.

Automation and Product Development Intern

2022–2024

Thoplo Machine Pvt. Ltd. & Baliyo Ventures Pvt. Ltd.

- Contributed to the mechanical design, fabrication, assembly, and testing of automated production machinery and supporting systems.
- Supported PLC lab setup and wiring, industrial equipment assembly, machining, fiber-reinforced molding, and production-focused fabrication.

SELECTED ENGINEERING PROJECTS

PlaiPin — Modular AI Companion Robot Platform

- Led mechanical CAD development of the core device and modular body configurations, creating assemblies that integrate actuation, sensors, and onboard hardware. [ADD MEASURABLE RESULT OR PROTOTYPE COUNT]

Mini TurtleBot — Low-Cost Modular Mobile Robot | ATR Lab

- Designed a chassis and motion architecture around low-cost fabrication, modular expansion, and straightforward integration of robotic hardware. [ADD TEST RESULT OR COST]

KAMI Head — Human-Robot Interaction Module

- Designed an expressive robotic head assembly combining mechanical motion systems and embedded components for human-robot interaction. [ADD DEGREES OF FREEDOM OR DEMO OUTCOME]

TECHNICAL SKILLS

Mechanical Design: Autodesk Inventor Professional, Fusion 360, SolidWorks, parametric assemblies, mechanisms, 3D modeling, bill-of-materials development

Robotics & Embedded: ROS, Arduino, microcontrollers, actuators, sensors, motor drivers, electromechanical integration

Fabrication: FDM and resin 3D printing, CNC machining, soldering, power tools, silicone, polyurethane, and fiber-reinforced molding

Automation: PLC fundamentals, industrial equipment assembly, wiring, and production-oriented fabrication

EDUCATION

B.S. in Mechatronics Engineering; Minor in Computer Science

Expected 2028

Kent State University — Kent, Ohio

Academic focus: Robotics, mechanical systems, electronics, prototyping, and applied engineering design.

LEADERSHIP & INTERNATIONAL ACTIVITIES

Founder & Organizer — Young Scientist Community, RECAST, Tribhuvan University

Founded and organized robotics education, technical workshops, STEM events, and community outreach. [ADD MEMBERS, EVENTS, OR STUDENTS REACHED]

Selected competitions & challenges: World Robot Olympiad (Thailand); FIRST Global Challenge (Switzerland); Asian Development Bank Hackathon; Robo-Battle Bots; Robo Soccer. [ADD ROLE, YEAR, AND RESULT]

Selected conferences & programs: Founders, Inc. (San Francisco); Asian Pacific Conference of Young Scientists. [ADD ROLE OR SELECTION]